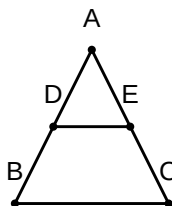


CBSE Class 10 Mathematics — Triangles — Solutions

1. 1. In $\triangle ABC$, if $DE \parallel BC$ such that $AD = 2$ cm, $DB = 3$ cm, and $AE = 4$ cm, find the length of EC .



Answer: (C)

Solution:

1. Since $DE \parallel BC$, by Thales Theorem (BPT), we have $\frac{AD}{DB} = \frac{AE}{EC}$.
2. Substitute the given values: $\frac{2}{3} = \frac{4}{EC}$.
3. Solve for EC : $2 \times EC = 12$, so $EC = 6$ cm.

Derivation:

$$\begin{aligned}\frac{AD}{DB} &= \frac{AE}{EC} \\ \frac{2}{3} &= \frac{4}{EC} \\ \Rightarrow 2 \times EC &= 12 \\ \Rightarrow EC &= 6\end{aligned}$$

Triangles — Basic Proportionality Theorem

2. 2. If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio $\frac{ar(\triangle ABC)}{ar(\triangle PQR)}$ is:

Answer: (A)

Solution:

1. According to the area of similar triangles theorem, the ratio of areas is equal to the square of the ratio of their corresponding sides.
2. The ratio of sides is $\frac{1}{3}$.
3. Square of the ratio is $(\frac{1}{3})^2 = \frac{1}{9}$.

Derivation:

$$\frac{ar(\triangle ABC)}{ar(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Triangles — Areas of Similar Triangles

3. **3.** Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer: (B)

Solution:

1. Let the poles be $AB = 6$ m and $CD = 11$ m. The distance between them is $BD = 12$ m.
2. Draw a line from A parallel to BD meeting CD at E .
3. Then $CE = CD - ED = 11 - 6 = 5$ m.
4. In right $\triangle AEC$, $AC^2 = AE^2 + CE^2 = 12^2 + 5^2 = 144 + 25 = 169$.
5. $AC = 13$ m.

Derivation:

$$AC = \sqrt{12^2 + (11 - 6)^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Triangles — Pythagoras Theorem

4. **4.** In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $AD = 3$ cm, $DB = 2$ cm, $AE = 4.5$ cm, and $EC = 3$ cm. Which of the following is true?

Answer: (A)

Solution:

1. Check if $\frac{AD}{DB} = \frac{AE}{EC}$.
2. $\frac{AD}{DB} = \frac{3}{2} = 1.5$.
3. $\frac{AE}{EC} = \frac{4.5}{3} = 1.5$.
4. Since the ratios are equal, $DE \parallel BC$ by the Converse of BPT.

Derivation:

$$\frac{AD}{DB} = \frac{3}{2} = 1.5 \text{ and } \frac{AE}{EC} = \frac{4.5}{3} = 1.5 \\ \Rightarrow DE \parallel BC$$

Triangles — Converse of BPT

5. **5.** If $\triangle ABC \sim \triangle DEF$, $\angle A = 50^\circ$, and $\angle E = 70^\circ$, then $\angle C$ is:

Answer: (C)

Solution:

1. Since $\triangle ABC \sim \triangle DEF$, corresponding angles are equal: $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.
2. Given $\angle E = 70^\circ$, so $\angle B = 70^\circ$.
3. In $\triangle ABC$, $\angle A + \angle B + \angle C = 180^\circ$.
4. $50^\circ + 70^\circ + \angle C = 180^\circ$.
5. $\angle C = 180^\circ - 120^\circ = 60^\circ$.

Derivation:

$$\angle C = 180^\circ - (\angle A + \angle B) = 180^\circ - (50^\circ + 70^\circ) = 60^\circ$$

Triangles — Similarity of Triangles

6. **6.** Assertion (A): If $\triangle ABC \sim \triangle DEF$ and $AB = 2$ cm, $DE = 4$ cm, then the ratio of their areas is $1 : 4$. Reason (R): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.

Answer: (A)

Solution:

1. State the theorem regarding the ratio of areas of similar triangles.
2. Calculate the square of the ratio of sides.
3. Compare with the given ratio of areas.

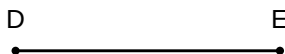
Derivation:

$$\text{Ratio of areas} = (AB/DE)^2 = (2/4)^2 = (1/2)^2 = 1/4 = 1 : 4$$

Since both Assertion and Reason are true and Reason explains the Assertion, option (A) is correct.

Triangles — Areas of Similar Triangles

7. **7.** Assertion (A): In $\triangle ABC$, if $DE \parallel BC$ with D on AB and E on AC , then $AD/DB = AE/EC$. Reason (R): Thales Theorem (Basic Proportionality Theorem) states that if a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.



Answer: (A)

Solution:

1. Identify the theorem being used.
2. Verify the statement of the Thales Theorem.
3. Check if the Assertion is the direct application of the Theorem.

Derivation:

Assertion is the mathematical statement of the Basic Proportionality Theorem.

Reason is the definition of the Basic Proportionality Theorem.

Both are true and R is the correct explanation.

Triangles — Basic Proportionality Theorem

8. **8.** Assertion (A): Two triangles are similar if their corresponding angles are equal.
Reason (R): If the corresponding sides of two triangles are in the same ratio, then the triangles are congruent.

Answer: (C)

Solution:

1. Check the validity of the AAA similarity criterion.
2. Check the validity of the SSS similarity vs congruence distinction.
3. Determine the truth value of both statements.

Derivation:

Assertion is true by AA similarity criterion.

Reason is false because proportional sides imply similarity, not necessarily congruence.

Therefore, A is true but R is false.

Triangles — Criteria for Similarity of Triangles

9. **9.** Assertion (A): In $\triangle ABC$, if $\angle A = 90^\circ$ and $AD \perp BC$, then $\triangle ADB \sim \triangle CDA$. Reason (R): Two triangles are similar if two angles of one triangle are respectively equal to two angles of the other triangle.

Answer: (A)

Solution:

1. Apply the AA similarity criterion for right triangles.
2. Verify the angle relationships.
3. Confirm the reasoning.

Derivation:

In $\triangle ADB$ and $\triangle CDA$, $\angle ADB = \angle CDA = 90^\circ$.

$\angle BAD = \angle ACD$ (both complement $\angle DAC$).

By AA similarity, $\triangle ADB \sim \triangle CDA$.

Reason is the AA criterion definition.

Triangles — Similarity of Triangles

10. **10.** Assertion (A): The perimeters of two similar triangles are in the same ratio as their corresponding sides. Reason (R): If $\triangle ABC \sim \triangle PQR$, then $AB/PQ = BC/QR = CA/RP = (AB + BC + CA)/(PQ + QR + RP)$.

Answer: (A)

Solution:

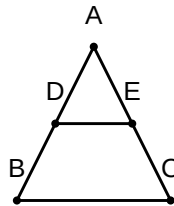
1. Use the properties of ratios.
2. Verify the perimeter formula.
3. Check the relationship between the assertion and the reason.

Derivation:

$$\begin{aligned}
 &\text{Given } \triangle ABC \sim \triangle PQR \\
 &AB/PQ = BC/QR = CA/RP = k. \\
 &AB = kPQ, BC = kQR, CA = kRP. \\
 &\text{Perimeter}(ABC) = AB + BC + CA = k(PQ + QR + RP) = k \times \text{Perimeter}(PQR). \\
 &\text{Ratio} = k. \\
 &\text{Both are true and R explains A.}
 \end{aligned}$$

Triangles — Similarity of Triangles

11. 11. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 3 \text{ cm}$, $DB = 4 \text{ cm}$, and $AE = 6 \text{ cm}$. Find the length of EC .



Answer:

8 cm

Solution:

1. Apply the Basic Proportionality Theorem (BPT) since $DE \parallel BC$.
2. Substitute the given values into $\frac{AD}{DB} = \frac{AE}{EC}$ and solve for EC .

Derivation:

$$\begin{aligned}
 \frac{AD}{DB} &= \frac{AE}{EC} \\
 \Rightarrow \frac{3}{4} &= \frac{6}{EC} \\
 \Rightarrow 3 \times EC &= 24 \\
 \Rightarrow EC &= 8 \text{ cm}
 \end{aligned}$$

Triangles — Basic Proportionality Theorem

- 12. 12.** Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If $Area(\triangle ABC) = 64 \text{ cm}^2$ and $Area(\triangle DEF) = 121 \text{ cm}^2$, and $EF = 11 \text{ cm}$, find the length of BC .

Answer:

$$8\text{cm}$$

Solution:

1. Use the theorem: The ratio of areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
2. Set up the equation $\frac{Area(\triangle ABC)}{Area(\triangle DEF)} = (\frac{BC}{EF})^2$ and solve for BC .

Derivation:

$$\begin{aligned}\frac{64}{121} &= (\frac{BC}{11})^2 \\ \Rightarrow \sqrt{\frac{64}{121}} &= \frac{BC}{11} \\ \Rightarrow \frac{8}{11} &= \frac{BC}{11} \\ \Rightarrow BC &= 8\text{cm}\end{aligned}$$

Triangles — Areas of Similar Triangles

- 13. 13.** In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PQR$. If $PS = 2 \text{ cm}$, $SQ = 3 \text{ cm}$, and $PT = 1.5 \text{ cm}$, find the length of TR .

Answer:

$$2.25\text{cm}$$

Solution:

1. Prove $\triangle PST \sim \triangle PQR$ by AA similarity criterion since $\angle P = \angle P$ and $\angle PST = \angle PQR$.
2. Use the proportionality ratio $\frac{PS}{PQ} = \frac{PT}{PR}$ where $PQ = PS + SQ$ and $PR = PT + TR$.

Derivation:

$$\begin{aligned}\frac{2}{2+3} &= \frac{1.5}{1.5+TR} \\ \Rightarrow \frac{2}{5} &= \frac{1.5}{1.5+TR} \\ \Rightarrow 3+2TR &= 7.5 \\ \Rightarrow 2TR &= 4.5 \\ \Rightarrow TR &= 2.25\text{cm}\end{aligned}$$

Triangles — Criteria for Similarity of Triangles

- 14. 14.** A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

Answer:

$$42\text{m}$$

Solution:

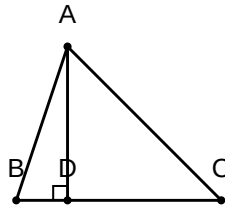
1. Recognize that the triangles formed by the objects and their shadows are similar due to the same sun angle.
2. Set up the proportion $\frac{Height_1}{Shadow_1} = \frac{Height_2}{Shadow_2}$ and solve for the tower height.

Derivation:

$$\begin{aligned}\frac{6}{4} &= \frac{h}{28} \\ \Rightarrow 4h &= 168 \\ \Rightarrow h &= 42m\end{aligned}$$

Triangles — Applications of Similar Triangles

- 15. 15.** In $\triangle ABC$, $AD \perp BC$. Prove that $AB^2 + CD^2 = AC^2 + BD^2$.



Answer:

Proof based on Pythagoras Theorem

Solution:

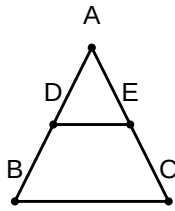
1. Apply Pythagoras theorem in $\triangle ABD$ ($AB^2 = AD^2 + BD^2$) and $\triangle ACD$ ($AC^2 = AD^2 + CD^2$).
2. Express AD^2 from both equations and equate them to show the result.

Derivation:

$$\begin{aligned}AB^2 - BD^2 &= AD^2 \\ AC^2 - CD^2 &= AD^2 \\ AB^2 - BD^2 &= AC^2 - CD^2 \\ AB^2 + CD^2 &= AC^2 + BD^2\end{aligned}$$

Triangles — Pythagoras Theorem

- 16. 16.** In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm, and $CE = (5x - 3)$ cm. Find the value of x .



Answer:

$$x = 1 \text{ or } x = 0.8$$

Solution:

1. Apply Thales Theorem (Basic Proportionality Theorem) since $DE \parallel BC$.
2. Set up the ratio $\frac{AD}{BD} = \frac{AE}{CE}$.
3. Solve the resulting quadratic equation for x .

Derivation:

$$\begin{aligned} \frac{4x - 3}{3x - 1} &= \frac{8x - 7}{5x - 3} \\ (4x - 3)(5x - 3) &= (8x - 7)(3x - 1) \\ 20x^2 - 12x - 15x + 9 &= 24x^2 - 8x - 21x + 7 \\ 20x^2 - 27x + 9 &= 24x^2 - 29x + 7 \\ 4x^2 - 2x - 2 &= 0 \\ 2x^2 - x - 1 &= 0 \\ (2x + 1)(x - 1) &= 0 \\ x = 1, x &= -0.5 \end{aligned}$$

Triangles — Basic Proportionality Theorem

- 17. 17.** Two poles of height 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.

Answer:

$$13 \text{ m}$$

Solution:

1. Represent the poles as vertical segments on a horizontal ground.
2. Form a right-angled triangle by drawing a line parallel to the ground from the top of the shorter pole.
3. Use the Pythagorean theorem to calculate the hypotenuse.

Derivation:

Let $AB = 6$ m and $CD = 11$ m be the poles.

Distance between feet $AC = 12$ m.

Draw $BE \perp CD$ such that $CE = AB = 6$ m.

$$DE = CD - CE = 11 - 6 = 5 \text{ m.}$$

In right $\triangle BED$, $BD^2 = BE^2 + DE^2$

$$BD^2 = 12^2 + 5^2 = 144 + 25 = 169$$

$$BD = \sqrt{169} = 13 \text{ m}$$

Triangles — Pythagoras Theorem

- 18. 18.** In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. If $AD = 8$ cm and $BD = 4$ cm, find the length of CD .

Answer:

16 cm

Solution:

1. Use the property of similarity in a right triangle where the altitude to the hypotenuse divides the triangle into two triangles similar to the whole and each other.
2. Establish the ratio $\triangle ADB \sim \triangle CDA$.
3. Solve for CD using $\frac{AD}{BD} = \frac{CD}{AD}$.

Derivation:

In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$.

$$\triangle ADB \sim \triangle CDA$$

$$\frac{AD}{CD} = \frac{BD}{AD}$$

$$AD^2 = BD \times CD$$

$$8^2 = 4 \times CD$$

$$64 = 4 \times CD$$

$$CD = 16 \text{ cm}$$

Triangles — Similarity of Triangles

- 19. 19.** The perimeters of two similar triangles are 30 cm and 20 cm respectively. If one side of the first triangle is 9 cm, find the length of the corresponding side of the second triangle.

Answer:

6 cm

Solution:

1. Recall that the ratio of perimeters of two similar triangles is equal to the ratio of their corresponding sides.
2. Set up the proportion $\frac{P_1}{P_2} = \frac{S_1}{S_2}$.
3. Substitute the given values and solve for S_2 .

Derivation:

$$\begin{aligned}\frac{P_1}{P_2} &= \frac{S_1}{S_2} \\ \frac{30}{20} &= \frac{9}{S_2} \\ 30 \times S_2 &= 20 \times 9 \\ 30 \times S_2 &= 180 \\ S_2 &= \frac{180}{30} = 6 \text{ cm}\end{aligned}$$

Triangles — Areas of Similar Triangles

- 20. 20.** In $\triangle ABC$, P and Q are points on sides AB and AC such that $PQ \parallel BC$. If $AP = 3$ cm, $PB = 2$ cm and $AQ = 2.7$ cm, find AC .

Q
•

Answer:

4.5 cm

Solution:

1. Apply the Thales theorem: $\frac{AP}{AB} = \frac{AQ}{AC}$.
2. Calculate $AB = AP + PB = 3 + 2 = 5$ cm.
3. Solve the ratio for AC .

Derivation:

$$\begin{aligned}\frac{AP}{AP + PB} &= \frac{AQ}{AC} \\ \frac{3}{3 + 2} &= \frac{2.7}{AC} \\ \frac{3}{5} &= \frac{2.7}{AC} \\ 3 \times AC &= 5 \times 2.7 \\ 3 \times AC &= 13.5 \\ AC &= 4.5 \text{ cm}\end{aligned}$$

Triangles — Basic Proportionality Theorem

- 21. 21.** Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Use this theorem to find the length of EC in $\triangle ABC$ where $DE \parallel BC$, $AD = 3$ cm, $DB = 4$ cm, and $AE = 6$ cm.



Answer:

$$EC = 8cm$$

Solution:

1. State Basic Proportionality Theorem (BPT).
2. Construct the proof by drawing altitudes and using area ratios.
3. Apply the theorem to the given values: $\frac{AD}{DB} = \frac{AE}{EC}$.
4. Substitute the given values: $\frac{3}{4} = \frac{6}{EC}$.
5. Solve for EC .

Derivation:

$$\begin{aligned}\frac{AD}{DB} &= \frac{AE}{EC} \\ \frac{3}{4} &= \frac{6}{EC} \\ 3 \times EC &= 6 \times 4 \\ 3 \times EC &= 24 \\ EC &= \frac{24}{3} \\ EC &= 8cm\end{aligned}$$

Triangles — Basic Proportionality Theorem

- 22.** In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$. Prove that $AD^2 = BD \times CD$. If $BD = 4$ cm and $CD = 9$ cm, calculate the length of AD .

Answer:

$$AD = 6cm$$

Solution:

1. Use the property of similar triangles: $\triangle ADB \sim \triangle CDA$.
2. Write the ratio of corresponding sides: $\frac{AD}{CD} = \frac{BD}{AD}$.
3. Cross multiply to get $AD^2 = BD \times CD$.
4. Substitute the given values $BD = 4$ and $CD = 9$.
5. Calculate $AD = \sqrt{36} = 6$ cm.

Derivation:

$$\triangle ADB \sim \triangle CDA$$

$$\frac{AD}{CD} = \frac{BD}{AD}$$

$$AD^2 = BD \times CD$$

$$AD^2 = 4 \times 9$$

$$AD^2 = 36$$

$$AD = 6\text{cm}$$

Triangles — Similarity of Triangles

- 23. 23.** Two poles of heights a meters and b meters are p meters apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ meters.

Answer:

$$\text{Height} = \frac{ab}{a+b}$$

Solution:

1. Let the poles be $AB = a$ and $CD = b$. Let them be p apart.
2. Let $EF = h$ be the height of the intersection point.
3. Show $\triangle EFC \sim \triangle ABC$ and $\triangle EFD \sim \triangle ABD$.
4. Express $\frac{EF}{AB} = \frac{FC}{BC}$ and $\frac{EF}{CD} = \frac{FD}{BD}$.
5. Sum the ratios to eliminate the segments of p and solve for h .

Derivation:

$$\frac{h}{a} = \frac{FC}{p} \text{ and } \frac{h}{b} = \frac{FD}{p}$$

$$\frac{h}{a} + \frac{h}{b} = \frac{FC + FD}{p}$$

$$h\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{p}{p} = 1$$

$$h\left(\frac{a+b}{ab}\right) = 1$$

$$h = \frac{ab}{a+b}$$

Triangles — Similarity and Ratios

- 24. 24.** Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Using this, find the ratio of the areas of two similar triangles if their sides are in the ratio 4 : 9.

Answer:

$$\text{Ratio} = 16 : 81$$

Solution:

1. State the theorem for the ratio of areas of similar triangles.
2. Draw two similar triangles and their altitudes.

3. Use similarity to show $\frac{Area_1}{Area_2} = (\frac{side_1}{side_2})^2$.
4. Apply the given ratio $\frac{s_1}{s_2} = \frac{4}{9}$.
5. Calculate the square of the ratio: $(\frac{4}{9})^2 = \frac{16}{81}$.

Derivation:

$$\begin{aligned}\frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= (\frac{AB}{PQ})^2 \\ \frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= (\frac{4}{9})^2 \\ \frac{ar(\triangle ABC)}{ar(\triangle PQR)} &= \frac{16}{81}\end{aligned}$$

Triangles — Areas of Similar Triangles